

openAWSEM Installation

- Code available at <https://github.com/cabb99/openawsem>

 README.md 

OpenAWSEM

An implementation of the AWSEM coarse-grained protein folding forcefield in OpenMM

The OpenAWSEM code is currently being tested. Use at your own risk. And let us know what you find. :-)

Installation

- Clone openawsem repository

```
git clone https://github.com/npschafer/openawsem.git
```

Instalación de open3SPN2

- Code available at <https://github.com/cabb99/open3spn2>
- Installation also available with conda
- `$ conda install -c wolynes-lab open3spn2`

The screenshot shows the GitHub repository page for `cabb99 / open3spn2`. The repository is described as "An implementation of the 3SPN2 and 3SPN2.C coarse-grained DNA forcefields in OpenMM". It has 100 commits, 1 branch, 0 packages, 2 releases, 2 contributors, and is licensed under MIT. The repository is currently on the `master` branch. A table of recent commits is shown below the repository information.

Commit	Description	Time
cabb99 Fixes docs		Latest commit 14091ed 2 hours ago
.github/ISSUE_TEMPLATE	Update issue templates	last month
Tests	Adds energy logs	4 months ago
docs	Changes requirements to conda	3 hours ago
examples	Adds protein-DNA example	4 hours ago
open3SPN2	Implements pip directory format	3 hours ago

Documentation

- <http://openawsem.org/>
- <https://open3spn2.readthedocs.io/>

Tutorial

“What I cannot create, I do not understand.” — Feynman

DNA from sequence

This section allows you to quickly simulate a piece of DNA.

First import the ff3SPN2 module.

```
import ff3SPN2
```

If you obtain a `ModuleNotFoundError` error then you may need to check that `ff3SPN2` is in the installation path as detailed in [Installation](#).

After importing the module, follow the next steps to create a DNA system and add the 3SPN2 forces. Make sure you have installed X3DNA before this step.

```
# Initialize the DNA from a sequence.  
# DNA type can be changed to 'A' or 'B'  
  
seq='ATACAAAGGTGCGAGGTTTCTATGCTCCCACG'  
dna=ff3SPN2.DNA.fromSequence(seq,dna_type='B_curved')
```

Simulate a protein

- Create the openawsem project

```
mm_create_project.py 1r69
```

```
cd 1r69
```

- Start with a short simulation and check that everything is working

```
./mm_run.py 1r69 --to rep0 --platform CPU --steps 1e3 --tempStart 500  
--tempEnd 200 -f forces_setup.py
```

- Visualize the trajectory using VMD
- Plot the Q value from the info.dat file

Fold a protein

Backup your previous project with a different name

```
mv 1r69 1r69_test
```

- Create an AWSEM project using fragment memory

```
mm_create_project.py 1r69 --frag --extended
```

- Change the file `setup_forces.py` to include the fragment memory

Uncomment this line:

```
# fragment_memory_term(oa, frag_file_list_file="./frags.mem", npy_frag_table="./frags.npy", UseSavedFragTable=True),
```

Comment this line:

```
fragment_memory_term(oa, frag_file_list_file="./single_frgs.mem", npy_frag_table="./single_frgs.npy",  
UseSavedFragTable=False),
```

- Run the simulation

```
mm_run.py extended --to rep0 --platform CPU --steps 1e5 --tempStart 500 --tempEnd 200 -f forces_setup.py
```

- Visualize the simulation using VMD
- Plot the Q value from the `info.dat` file

Simulate a DNA-protein complex

- Create a new folder

```
mkdir 1lmb
```

```
cd 1lmb
```

- Download a protein-DNA complex structure from the PDB

```
wget https://files.rcsb.org/download/1LMB.pdb 1lmb
```

- Open a python kernel (or jupyter notebook)

```
ipython
```

- Follow the instructions from the following link

https://github.com/cabb99/open3spn2/blob/master/examples/Protein_DNA/Protein_DNA_example_CPU.ipynb

- Change the following line

```
dna=open3SPN2.DNA.fromCoarsePDB('clean.pdb')
```

```
dna.computeTopology(template_from_X3DNA=True)
```

To:

```
dna=open3SPN2.DNA.fromCoarsePDB('clean.pdb', template_from_X3DNA=False)
```

- Visualize the simulation using VMD